

PVsyst - Simulation report

Standalone system

Project: Kermanshad - Charge station

Variant: New simulation variant

Standalone system with batteries

System power: 7.40 kWp

Kermanshah - charge station (parking-kut) - Iran



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VC4, Simulation date:
12/07/26 12:31
with V8.1.3

Project summary

Geographical Site

Kermanshah - charge station (parking-kut)
Iran

Situation

Latitude 34.3242 °(N)
Longitude 47.0945 °(E)
Altitude 1329 m
Time zone UTC+3.5

Project settings

Albedo 0.20

Weather data

Kermanshah - charge station (parking-kut)
Meteonorm 8.2 (1985-2002), Sat=100% - Synthetic

System summary

Standalone system

Orientation #1

Fixed plane

Tilt/Azimuth 33.2 / 0 °

Standalone system with batteries

Near Shadings

Linear shadings : Fast (table)

User's needs

Daily household consumers
Constant over the year
Average 33.5 kWh/Day

System information

PV Array

Nb. of modules 20 units
Pnom total 7.40 kWp

Battery pack

Technology Lithium-ion, NCA
Nb. of units 3 units
Voltage 50 V
Capacity 804 Ah

Results summary

Simulation step	Hourly				
Useful energy from solar	9536.2 kWh/year	Specific production	1289 kWh/kWp/year	Perf. Ratio PR	53.19 %
Missing Energy	2696.8 kWh/year	Available solar energy	10576.6 kWh/year	Solar Fraction SF	77.83 %
Excess (unused)	465.0 kWh/year				

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General parameters

Standalone system

Simulation step

Hourly

Orientation #1

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User's needs

Daily household consumers

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Average 33.5 kWh/Day

Standalone system with batteries

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Near Shadings

Linear shadings : Fast (table)

PV Array Characteristics

PV module

Manufacturer Almaden

Model SEAB72T-370

(Original PVsyst database)

Unit Nom. Power 370 Wp

Number of PV modules 20 units

Nominal (STC) 7.40 kWp

Modules 2 units x 10 In series

At operating cond. (50°C)

Pmpp 6.67 kWp

U mpp 358 V

I mpp 19 A

Total PV power

Nominal (STC) 7 kWp

Total 20 modules

Module area 39.1 m²

Cell area 35.4 m²

Controller

Universal controller

Technology MPPT converter

Temp coeff. -5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 97.0 / 95.0 %

Battery Management control

Threshold commands as SOC calculation

Charging SOC = 0.96 / 0.80

Discharging SOC = 0.10 / 0.35

Battery Storage

Battery

Manufacturer Tesla

Model Powerwall2

Technology Lithium-ion, NCA

Nb. of units 3 in parallel

Discharging min. SOC 10.0 %

Stored energy 36.5 kWh

Battery Pack Characteristics

Voltage 50 V

Nominal Capacity 804 Ah (C10)

Temperature External ambient temperature



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Array losses

Array Soiling Losses

Average loss Fraction 30.0 %

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
2.5%	7.5%	12.5%	17.5%	22.5%	27.5%	32.5%	37.5%	42.5%	47.5%	52.5%	57.5%

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 27.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 318 mΩ
Loss Fraction 1.50 % at STC

Serie Diode Loss

Voltage drop 0.7 V
Loss Fraction 0.2 % at STC

LID - Light Induced Degradation

Loss Fraction 2.00 %

Module Quality Loss

Loss Fraction -0.35 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.05 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

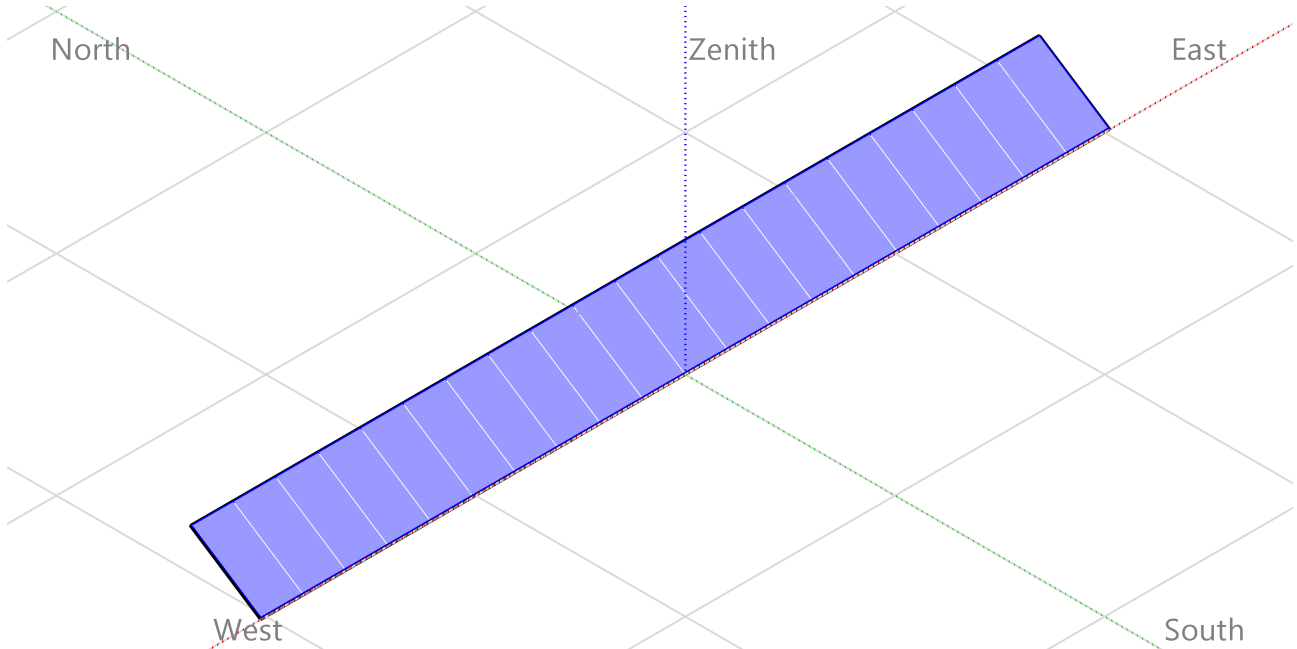


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Near shadings parameter

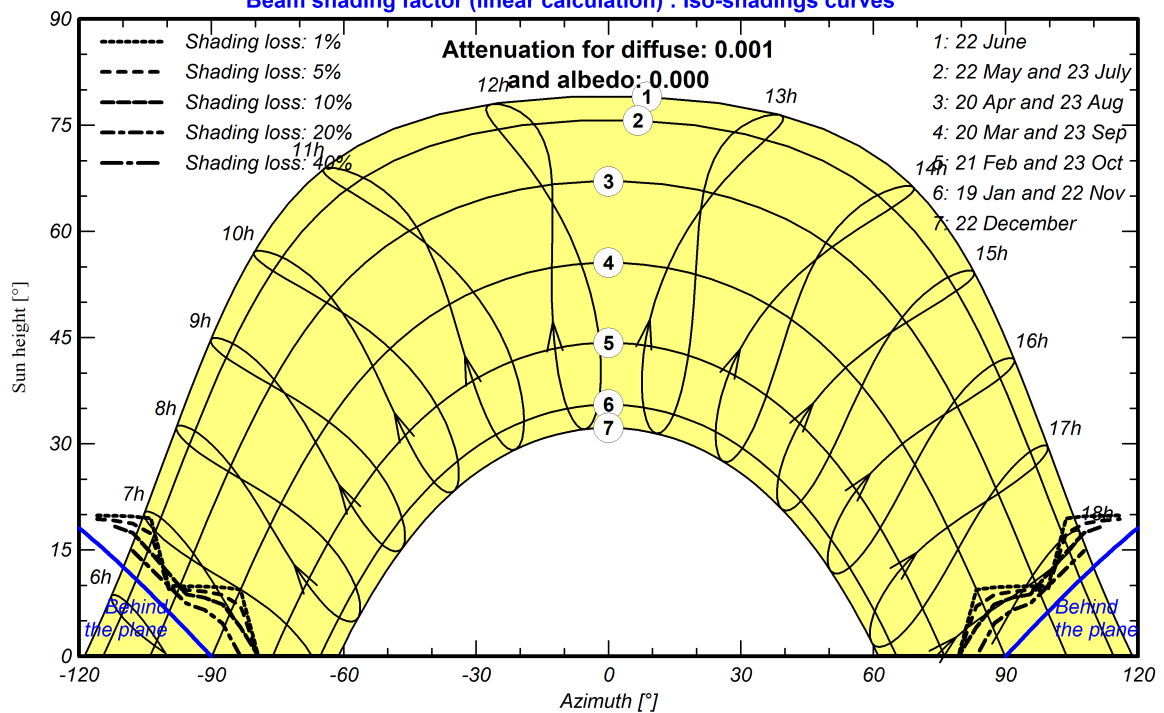
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 33.2°/ 0°

Beam shading factor (linear calculation) : Iso-shadings curves





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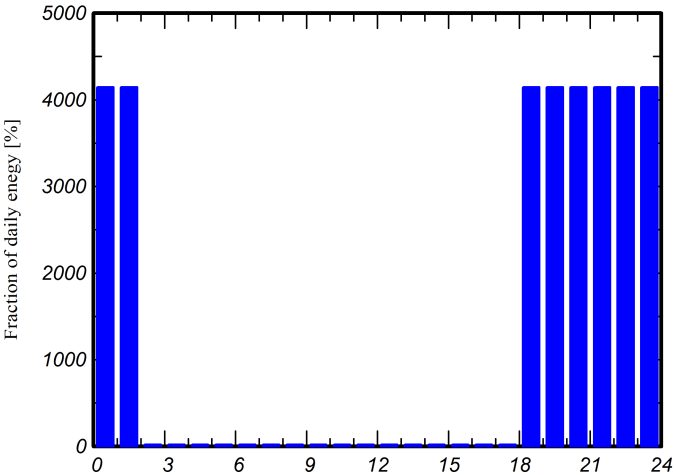
Detailed User's needs

Daily household consumers, Constant over the year, average = 33.5 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
golf carts	4	1031/app	8.0	32992
golf carts charger	4	5/app	24.0	499
Stand-by consumers			24.0	24
Total daily energy				33515

Hourly distribution





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Main results

Simulation step

Hourly

System Production

Useful energy from solar 9536.2 kWh/year
Available solar energy 10576.6 kWh/year
Excess (unused) 465.0 kWh/year

Perf. Ratio PR 53.19 %
Solar Fraction SF 77.83 %

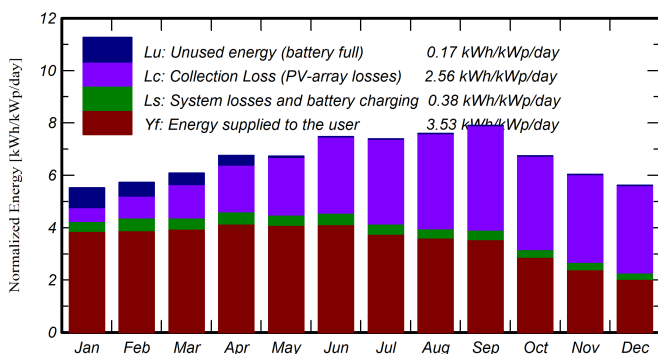
Loss of Load

Time Fraction 7.4 %
Missing Energy 2696.8 kWh/year

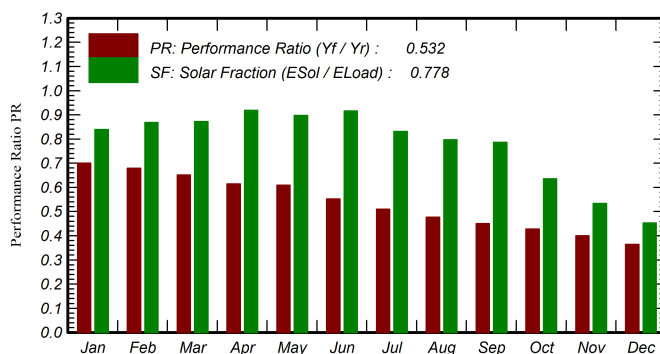
Battery ageing (State of Wear)

Cycles SOW 70.7 %
Static SOW 88.6 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	102.7	165.0	1109	171.9	151	888	1039	0.839
February	113.6	146.5	979	106.9	131	808	938	0.868
March	155.5	161.9	1064	98.4	131	908	1039	0.872
April	191.5	163.9	1060	78.0	84	922	1005	0.919
May	220.9	158.0	998	9.8	97	942	1039	0.898
June	252.7	158.7	973	0.0	88	917	1005	0.916
July	250.7	151.1	911	0.0	174	865	1039	0.831
August	233.0	144.1	874	0.0	209	830	1039	0.797
September	202.1	134.0	836	0.0	214	791	1005	0.787
October	152.5	108.2	700	0.0	378	661	1039	0.636
November	112.3	84.9	571	0.0	470	535	1005	0.534
December	98.2	73.1	502	0.0	570	469	1039	0.453
Year	2085.8	1649.4	10577	465.0	2697	9536	12233	0.778

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



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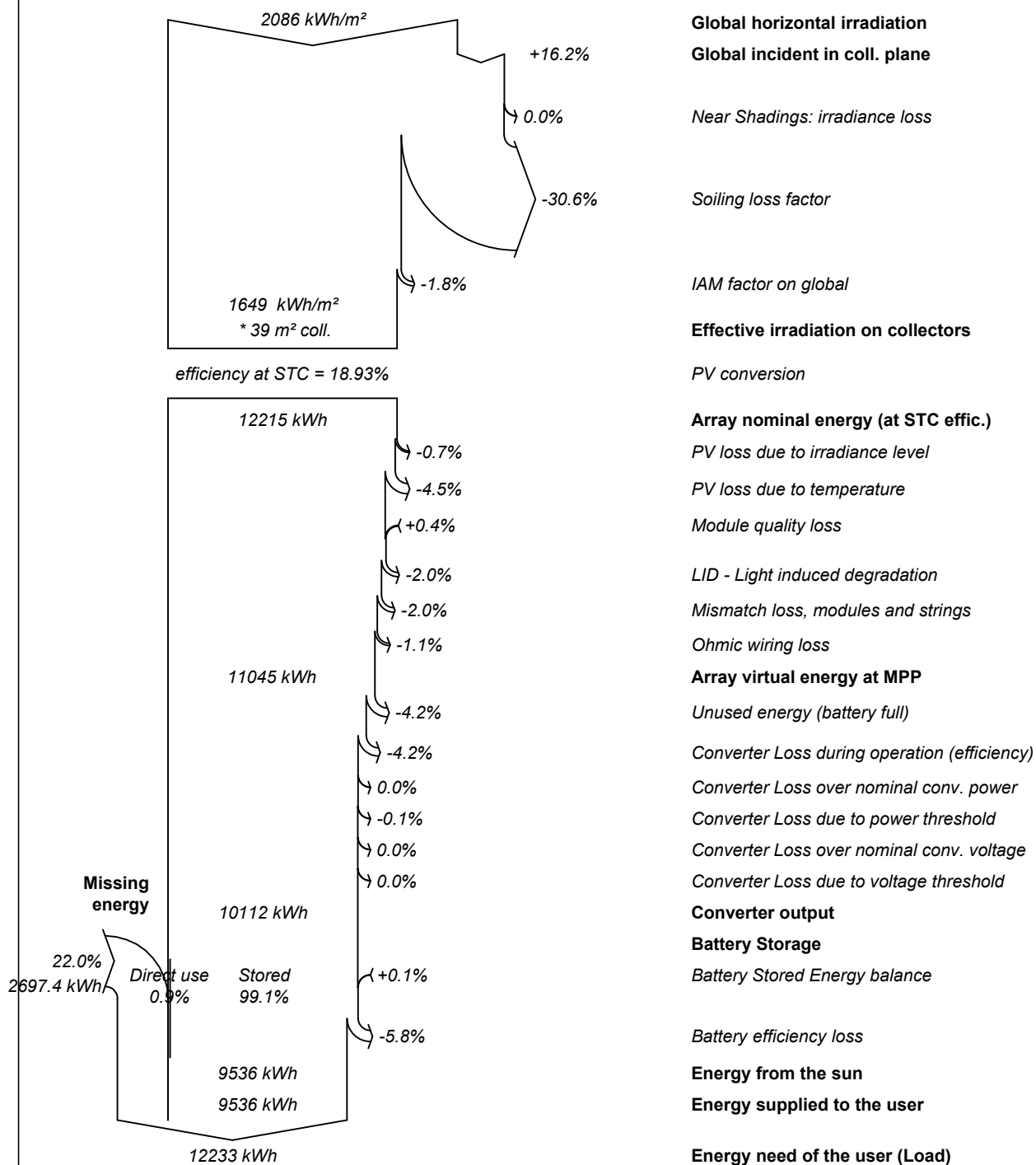
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Loss diagram





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Predef. graphs

Daily Input/Output diagram

